# Reading, Not Reasoning: A Causal Faithfulness Audit of Chain-of-Thought in Distilled Chart-and-Table VLMs

Anonymous Authors*
Anonymous Institution
anonymous

**Abstract**

Many multimodal systems compress agentic visual reasoning into a single tool-free VLM forward pass via chain-of-thought supervised fine-tuning. We ask what such distillation actually transfers. Across chart, table, natural-image counting, and rendered financial QA, CoT SFT reliably improves accuracy—often strongly—yet a causal faithfulness audit shows that the emitted written chain is not load-bearing: corrupting an on-path numeric intermediate flips the final answer no more often than a format-matched control, and the model almost never follows the injected value, instead snapping back to the answer readable from the original evidence. The information source behind each gain is task-dependent: models re-read visible charts and tables, reconstruct natural-image counts from redundant enumerations in the rationale, or copy the chain's own conclusion. Even a shortcut-removing curriculum with gold-program-verified operands raises accuracy substantially without installing load-bearing intermediate computation. Accuracy-only internalization claims conflate reading with reasoning; causal probes over emitted rationales can separate the two.

## 1 Introduction

Agentic VLM systems—iterative frame retrieval, self-reflection, critic-orchestrated re-reading, visual tool use—often improve accuracy, but their inference-time tools are expensive. A natural compression step is to convert an expert trajectory into a written rationale or latent trace and fine-tune a VLM to execute one forward pass at test time (Adhikari and Lapata, 2026; Carbune et al., 2024; Hu et al., 2024; Tao et al., 2026; Wei et al., 2026). Accuracy improvements from such distillation are routinely reported, but a prior question is left unasked: when distillation lifts accuracy, does the student's emitted CoT carry the causal work, or does it merely record, paraphrase, or reveal an answer found elsewhere?

We address this question directly. We assess causal faithfulness by corrupting emitted numeric intermediates and comparing against format-matched controls (Lanham et al., 2023). The causal claim is about the emitted written chain; the probes do not rule out hidden-state or latent computation that never surfaces in the rationale.

**Contributions.**

1. We show that CoT SFT reliably and substantially improves accuracy on chart, table, natural-image counting, and rendered financial QA—establishing that there is a real accuracy signal for the faithfulness audit to interrogate.

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2. We demonstrate that accuracy gains and causal faithfulness decouple: a causal probe battery across four task types finds that the emitted rationale is not load-bearing, even when accuracy improvements are large and statistically robust.
3. We characterize the task-dependent information source behind each gain—re-reading visible evidence, exploiting redundant enumerations, or copying the chain's conclusion—and show these are distinguishable by the probe design.
4. We provide a pre-specified test showing that shortcut-removing curricula with gold-program-verified operands raise accuracy without installing load-bearing intermediate computation, ruling out a straightforward training-data explanation for the faithfulness null.

## 2 Related Work

**Internalizing visual reasoning.** Visual Program Distillation (Hu et al., 2024), chart-specific rationale transfer (Carbune et al., 2024; Chen et al., 2025), latent trajectory learning (Adhikari and Lapata, 2026; Li et al., 2026), and region-to-image or local-cue distillation (Tao et al., 2026; Wei et al., 2026) all aim to move work that previously required tools, crops, or expert traces into a cheaper model or inference path. These papers mainly validate the transfer by answer accuracy, sometimes with attention or qualitative trace analysis. Our question is narrower: after a no-tool VLM emits a discrete rationale, is the emitted intermediate value an answer-carrying state, or is it an artifact of a successful read?

**CoT faithfulness and visual-agent faithfulness.** Text-LLM faithfulness work shows that rationale scaffolds can be made more explicitly faithful (Lyu et al., 2023), that plausible rationales can omit true decision drivers (Turpin et al., 2023), that larger models can produce less answer-dependent CoT (Lanham et al., 2023), and that filler tokens can support hidden computation independent of human-readable steps (Pfau et al., 2024). Two critiques matter for our design: unlearning or deletion can measure contextual robustness rather than load-bearing computation (Tutek et al., 2025), and multiple-choice option bias can make some faithfulness metrics look like disguised accuracy (Bentham et al., 2024). In multimodal settings, recent audits target different objects: visual-thinking agents may ignore visual evidence (Liu et al., 2025b), CodeV perturbs tool-output images rather than internalized CoT tokens (Hou et al., 2026), and latent imagination work studies continuous intermediate states (Li et al., 2026). We therefore position our audit by object and mechanism: discrete CoT emitted by a distilled, single-forward VLM, evaluated with controls that separate re-reading, redundancy, and conclusion copying.

**Why charts and tables.** Chart and table QA are useful because the same task mixes visual extraction and numerical reasoning. ChartQA (Masry et al., 2022), TabMWP (Lu et al., 2023), CharXiv (Wang et al., 2024), ChartMuseum (Tang et al., 2025), chart-noise stress tests (Shin et al., 2025), and perception-bottleneck analyses (Liu et al., 2025a) all point to a persistent ambiguity: an accuracy improvement can come from reading values better, computing with them better, or both. Our experiments exploit that ambiguity instead of hiding it. By masking images, falsifying on-path operands, and comparing chart/table with natural-image counting, we ask which representation the answer actually reads from at the final step.

## 3 Experimental Setup

**Tasks.** *ChartQA* (Masry et al., 2022) (400 held-out open numeric questions, image-disjoint from SFT train images), *TabMWP* (Lu et al., 2023) (400 rendered-table, free-response numeric cases), TallyQA-complex natural-image counting (400 open integer questions), and FinQA rendered tables for the pre-specified curriculum probe. All four tasks fix the visual evidence in the input, making the causal question well-posed: does the written chain carry the computation that produces any SFT accuracy gain?

**Models.** Qwen3-VL 8B and 32B (dense) (Bai et al., 2025). All models are served locally.

**SFT pipeline.** For ChartQA and TabMWP, a Qwen3-VL-32B teacher generates step-by-step rationales on training images; we keep only traces whose final answer matches the gold answer under the numeric grader below. This yields 150 ChartQA traces and 160 TabMWP traces. The 8B students are LoRA-SFT models with the vision tower frozen, rank 16, $\alpha$=32, dropout .05, bf16, learning rate $10^{-4}$, per-device batch size 1, gradient accumulation 8, cosine schedule, and prompt tokens masked to $-100$ so supervision applies only to the assistant rationale and `ANSWER` line. The ChartQA/TabMWP 8B runs use two epochs, corresponding to 38 and 40 optimizer updates. The 32B students use the same LoRA target modules under NF4 QLoRA and run five epochs with greedy held-out evaluation after each epoch; the selected checkpoints are epoch 1 for ChartQA and epoch 4 for TabMWP. Dense 8B controls use full-SFT checkpoints with the vision tower, embeddings, and first three language layers frozen, leaving about 79.7% of parameters trainable. The FinQA curriculum arms use the same LoRA recipe for three epochs: 180/180 gold-program strict/vanilla traces, and 173/173 answer-conditioned text-teacher strict/vanilla traces. All answer generation and probe continuations use greedy decoding (`do_sample=False`; answer max-new 320–384 tokens, continuation max-new 64 tokens).

**Statistics.** McNemar exact tests for paired SFT deltas, using only discordant pairs; Holm correction across the four headline ChartQA/TabMWP cells leaves all tests below .01. At $n$=400, minimum detectable net is about 8–9 points, and the observed paired SFT gains have 66 and 46 discordant pairs at 8B/32B.

**Robustness to known confounds.** We do not claim the CORRUPT intervention itself is new; it is useful only as part of a controlled battery. To address the "contextual faithfulness" concern (Tutek et al., 2025), we compare CORRUPT against SHUFFLE, paraphrase, filler-token, deletion, truncation, image-mask, and a snap/follow/other readout that asks whether the final answer follows the injected value. To address disguised-accuracy and option-bias concerns (Bentham et al., 2024), the main faithfulness substrates are free-response numeric QA rather than multiple choice. To avoid reporting a probe artifact as mechanism, we treat a positive corruption flip as insufficient by itself; the headline evidence is convergence across controls and regimes.

**Probe construction.** For each task/model cell, we first ask the evaluated model to solve the original visual question with the image/table visible and the prompt `Solve step by step, end with 'ANSWER: <final>'`. The text before the first `ANSWER:` is the emitted CoT. A case is probe-eligible only if this CoT is nonempty and the extracted final answer is correct. The generic chart/table battery then edits that model-owned CoT with a fixed edit seed `Random(0)`: CORRUPT samples one numeric token from the CoT and replaces value $v$ by $2v + 7$ (integers remain integers; decimals are

| |
|---|
| **Original CoT:** read value $a$, read value $b$, compute $a - b = c$, answer $c$. |
| **CORRUPT:** read value $a$, read value $b$, compute $a - b = \tilde{c}$, answer ? |
| **SHUFFLE:** same rationale tokens, shuffled order, answer ? |
| **Readout:** snap to true $c$ / follow injected $\tilde{c}$ / other. |

**Figure 1: Probe schematic.** CORRUPT tests whether a falsified written intermediate is load-bearing; SHUFFLE is a format control with no injected target value.

rendered to one decimal), while SHUFFLE shuffles sentence order without injecting a target value. Additional controls paraphrase the CoT with an exact number-multiset guard, replace tokens with length-matched filler, truncate early sentences, or delete the last number-bearing steps. The FinQA targeted probe is stricter: it uses the gold program to choose a table operand that is non-constant, appears in the student's CoT, and whose replacement by $2v + 7$ changes the gold-program final answer; it then tests both operand-only corruption and a consistently propagated corrupted chain. Each edited CoT is force-continued with the prompt "Reasoning so far: ... Given ONLY the reasoning above, state the final answer now as `ANSWER: <final>`". The masked condition drops the image/table from this force-continue call while keeping the same edited text.

**Grading and readout.** Open numeric answers are graded in the task's original answer units. We strip commas and percent signs from the gold answer, extract all numeric strings from the prediction, and mark a match if any predicted number is within $0.05|g| + 10^{-6}$ of gold value $g$; nonnumeric gold answers fall back to normalized substring matching. Rounding and redundant units are therefore tolerated only through the numeric value; symbolic equivalents that do not surface a matching number, empty outputs, and parse failures are incorrect. "Flip" means the forced final answer no longer matches the unedited model answer. For CORRUPT and FinQA targeted probes, the three-way readout is applied in priority order: snap if the answer matches the gold answer, follow if it does not snap but matches the injected wrong value (or injected gold-program final), and other otherwise. Boundary examples under 5% tolerance are: gold 100, injected 207, answer "about 96" → snap; answer "207" → follow; answer "200" → follow; answer containing both 100 and 207 → snap by precedence; and a SHUFFLE answer has no injected target, so off-gold responses are counted as other rather than follow.

## 4 Result I: CoT SFT Improves Accuracy

On the chart and table tasks, we run the actual distillation. A Qwen3-VL teacher generates step-by-step rationales; answer-consistency filtering keeps high-quality traces; LoRA/QLoRA SFT trains a single-forward student. All reported evaluation sets are image-disjoint from training and use open numeric grading. These gains are best read as improved single-pass readout unless the causal probe shows that the emitted chain itself carries the answer.

- **ChartQA,** $n$=400**:** 8B .713 → **.778**, net +6.5 points, CI [+2.5, +10.5], McNemar $p$=0.0021; 32B .725 → **.795**, net +7.0 points, CI [+3.8, +10.3], McNemar $p$=6.9×$10^{-5}$. The original 32B $n$=60 pilot had only four discordant pairs; the expanded evaluation has 37 gains vs. 9 losses.
- **TabMWP,** $n$=400**:** 8B .865 → **.963**, net +9.8 points, McNemar $p$=2.9×$10^{-9}$; 32B .835 → **.973**, net +13.8 points, McNemar $p$=2.1×$10^{-12}$.

Thus the paper is not a null-result critique of SFT. The accuracy gains are real, paired, and

**Table 1:** Causal CoT-intermediate probe on chart/table SFT cells. "Flip" = final answer changes after the intervention. "Snap" = answer returns to the true chart/table value after CORRUPT; "follow" = answer follows the injected value. A load-bearing chain requires CORRUPT > controls and nontrivial follow.

| Dataset/model | Condition | $n$ | CORRUPT | SHUFFLE | snap | follow |
|---|---|---|---|---|---|---|
| ChartQA 8B | present | 321 | .212 | .302 | .816 | .031 |
| ChartQA 32B | present | 318 | .051 | .047 | .981 | .016 |
| TabMWP 8B | present | 385 | .034 | .145 | .971 | .008 |
| TabMWP 32B | present | 389 | .033 | .072 | .969 | .015 |
| TabMWP 8B | masked | 385 | .278 | .299 | .722 | .088 |
| TabMWP 32B | masked | 389 | .075 | .103 | .928 | .039 |

adequately powered on static chart/table substrates. The remaining question is what mechanism those gains bought.

**Dense 8B control.** To rule out a low-rank-capacity explanation, we repeat the 8B arms with dense/full-SFT controls (vision tower, embeddings, and the first three language layers frozen; about 79.7% trainable parameters for ChartQA/TabMWP). The accuracy deltas remain positive: ChartQA .713→.763 (net +5.0 points, 39 gains/19 losses, $p$=0.0126) and TabMWP .865→.968 (net +10.3 points, 42 gains/1 loss, $p=1.1\times10^{-9}$). Available dense probes keep the LoRA signature: ChartQA's completed six-intervention battery has present-image CORRUPT/SHUFFLE = .100/.259, snap .917 and follow .030; TabMWP's completed core battery has present-table CORRUPT/SHUFFLE = .049/.161, snap .971 and follow .016; FinQA dense targeted arms keep operand-follow near zero while consistent-conclusion follow remains .947–.982. The remaining dense-control gap is the full six-intervention TabMWP present/masked battery, so the dense control is evidence against the LoRA-capacity confound but not yet the final multi-cell table.

## 5 Result II: The Causal Probe—the Written Chain Is Not Load-Bearing

For every correctly answered case with an emitted CoT, we intervene on the rationale and force-continue to a final answer. If a numeric intermediate carries the computation, CORRUPT should flip more often than SHUFFLE, and the final answer should follow the injected value rather than snap back to the original truth. Table 1 reports the high-power ChartQA and TabMWP cells.

Four findings follow.

1. **ChartQA and TabMWP agree.** In every present-image/table cell, CORRUPT $\leq$ SHUFFLE up to sampling noise. The model usually snaps back to the true value and almost never follows the injected value. TabMWP is the cleanest case: despite large SFT gains, follow is .008/.015 at 8B/32B.
2. **The gain subset itself is reading.** On TabMWP cases newly solved by SFT, corrupting the CoT flips 0/40 8B gains and 1/57 32B gains; follow is 1/40 at 8B (95% upper bound 12.9%) and 0/57 at 32B (95% upper bound 6.3%). The new correct answers are not being computed from the falsified written arithmetic.
3. **Masking diagnoses the information source.** On tables, masking the image raises flip and "other" errors sharply, especially for 8B, but follow remains small. When the table is visible, the

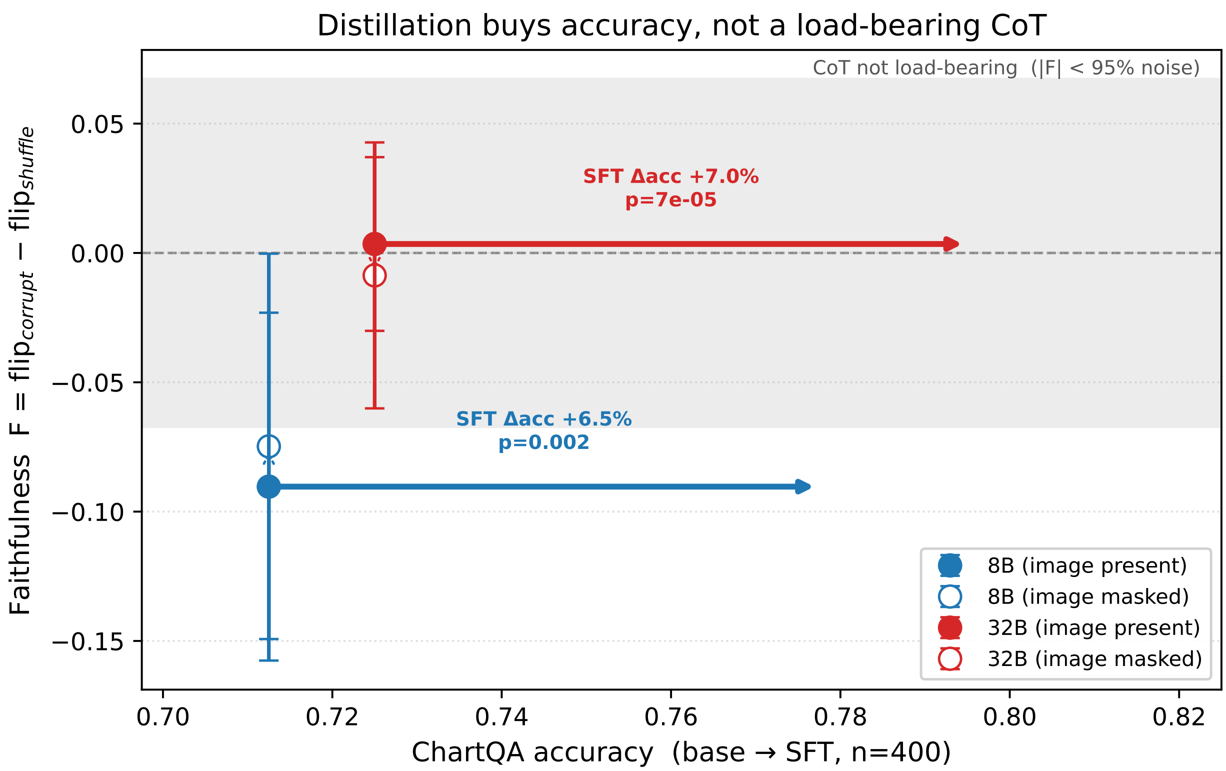


**Figure 2: Accuracy $\perp$ written-chain faithfulness.** CoT distillation lifts ChartQA accuracy by +6.5–7.0% (McNemar $p \leq .002$; horizontal arrows), yet the causal CoT metric stays on the $F \approx 0$ band ($F = \text{flip}_{\text{CORRUPT}} - \text{flip}_{\text{SHUFFLE}}$): corrupting a numeric intermediate flips the answer no more than shuffling the CoT. This is a conservative criterion: SHUFFLE already disrupts format without injecting any target value, so $F \leq 0$ means the falsified intermediate contributes no extra flip beyond pure format noise. Hollow markers show image-masked controls. $F$ is measured on the same SFT student used for the corresponding probe; error bars are 95% binomial CIs. Regenerated from the result store by `scripts/build_faithfulness_axis.py`.

final answer step re-reads and recomputes from the table; when the table is removed, the model often fabricates rather than faithfully using the bad chain.

4. **Other interventions point the same way.** Paraphrase behaves like CORRUPT (e.g., 32B ChartQA present .053 vs. .051), while filler/truncation/deletion and SHUFFLE mainly test format and token-budget sensitivity. Because SHUFFLE injects no target value, its paper-facing readout is flip/snap/other rather than follow. Existing summaries show this is a harsh control rather than an injected-value test: present-image/table SHUFFLE accuracy-after is .735/.987 on ChartQA 8B/32B and .857/.936 on TabMWP 8B/32B. The conclusion is carried by the whole battery, not by a single corruption statistic.

## 6 Result III: Task-Dependent Information Sources and a Constructive Test

**Natural-image counting.** To test whether the chart/table result is a chart artifact, we run a pre-specified TallyQA-complex counting probe on base Qwen3-VL-8B/32B, with no chart adapter. This is a genuinely perception-bottlenecked natural-image regime (free-form accuracy .415/.448).

Yet the emitted CoT's numeric content is again not load-bearing: across scales, follow is 0/345 with the image present and 1/345 when masked. Even when CORRUPT hits the count token itself, follow is 0/103 present and 1/103 masked. The model reconstructs the answer from redundant enumeration in the text, so masking the image is almost inert. This is the same reading-not-reasoning pattern with a different readable source: not the image, but the redundant chain.

**Shortcut-removing curriculum.** As a constructive counter-test, we train on rendered FinQA tables filtered to require multi-step, multi-cell arithmetic with a gold-program-verified flippable operand. The curriculum uses a deterministic gold-program teacher and a matched vanilla control; a text-teacher arm repeats the test with fluent answer-conditioned rationales. Accuracy rises substantially (8B: .080 base → .582 vanilla → .674 curriculum; 32B: .123 → .659 → .682). But the targeted probe still finds no load-bearing operand use: changing a verified on-path *input* operand yields follow 0/172 for the 8B curriculum and 0/175 for the 32B curriculum. When the chain's final conclusion is consistently rewritten to the recomputed value, the model follows it in 96–97% of cases. The mechanism is conclusion copying, not recomputation from operands. Thus even a shortcut-removing curriculum can improve accuracy without installing a load-bearing written chain.

## 7 Discussion and Limitations

**Implications.** (i) Claims of "internalizing agentic reasoning" need causal audits. Significant accuracy gains are compatible with re-reading visual evidence, reconstructing from redundant rationale text, or copying the rationale's conclusion. (ii) The load-bearing locus is task-dependent: answer-time evidence can live in pixels, in redundant text, or in a conclusion line. A single CoT faithfulness score without information-source controls is under-specified.

**Limitations.** All causal claims are about emitted rationales, not a proof that the model lacks latent internal computation. The chart/table and FinQA operands are input-readable by design, which is exactly why these substrates can distinguish re-reading from written-chain use; the scope does not include tasks whose required operands are unavailable in the input. The 32B students use QLoRA/NF4 rather than full bf16 fine-tuning. The main teacher family is Qwen3-VL, although the FinQA text-teacher arm reduces a format-specific teacher concern. Corrupting one numeric token is a sensitivity lower bound; we therefore report it only with shuffle/paraphrase/filler/truncation, image masking, snap/follow/other classification, and targeted gold-program operand tests.

## 8 Conclusion

We set out to compress tool-scaffolded visual reasoning into one forward pass and to verify what moved. The answer is not a blanket dismissal of CoT or of distillation: SFT works, often strongly. The causal result is more specific. In these chart, table, natural-counting, and rendered-FinQA regimes, the emitted answer reads from the easiest available source: pixels, redundant enumerations, or the chain's final conclusion. Accuracy can move while load-bearing written-chain faithfulness does not. That is the audit lesson behind *Reading, Not Reasoning.*

## Reproducibility Statement

All numbers regenerate from a fingerprinted, append-only result store (`results.jsonl`) via the released scripts; SFT, merge, battery, curriculum, and probe scripts are included. ChartQA/TabMWP builders, power table, faithfulness figures, N3 natural-counting probe, and N1 targeted curriculum probe each correspond to one script with fixed seeds or saved artifacts.

# A Artifact Map

Table 2 gives the minimum artifact map needed to reconstruct each headline table or figure without reading source code first. Paths are relative to the repository root. JSON hashes are 12-character SHA-256 prefixes of the files used in this draft.

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**Table 2: Primary artifacts for the SFT and probe results.** Probe edits use greedy decoding and fixed edit seed `Random(0)` unless otherwise noted; result-store rows also carry per-run fingerprints over split, method, decoding, seed, prompt, and code version.

| Result family | Result JSON / store | Script and seed | Checkpoint or input path |
|---|---|---|---|
| Global result store | `data/distill/results/results.jsonl`, SHA `31ef62ccb77a` | `scripts/regen_tables.py`; stored run seed/fingerprint | append-only store used by tables and map figures |
| ChartQA LoRA SFT accuracy | `data/distill/poc/lora_8b_chartqa/eval_n400.json`, SHA `2ce4d9175235`; `data/distill/poc/lora_32b_chartqa/eval_n400.json`, SHA `92e59e633a71` | `scripts/poc_sft.py`; `scripts/poc_sft_32b_qlora.py`; `scripts/eval_sft_n400.py` | train `data/distill/poc/chartqa_cot_train.jsonl`; adapters `data/distill/poc/lora_8b_chartqa`, `data/distill/poc/lora_32b_chartqa/epoch_1` |
| TabMWP LoRA SFT accuracy | `data/distill/poc/eval_sft_8b_tabmwp_n400.json`, SHA `95aaba7b47df`; `data/distill/poc/eval_sft_32b_tabmwp_n400.json`, SHA `b08953be8f1f` | `scripts/run_wu3_stage2.sh`; `scripts/eval_sft_n400.py` | train `data/distill/poc/tabmwp_cot_train.jsonl`; adapters `data/distill/poc/lora_8b_tabmwp`, `data/distill/poc/lora_32b_tabmwp/epoch_4` |
| ChartQA faithfulness battery | `data/distill/poc/battery_8b_present.json`, SHA `0ff0da834fc9`; `data/distill/poc/battery_32b_present.json`, SHA `34a57e280224`; masked companions in same directory | `scripts/battery_n400.py`, edit seed 0 | same ChartQA adapters; test split `data/distill/chartqa/test_cases_400.jsonl` |
| TabMWP faithfulness battery | `data/distill/poc/battery_tabmwp8b_present.json`, SHA `c27d11005883`; `data/distill/poc/battery_tabmwp32b_present.json`, SHA `078a37c4e4c6`; masked companions in same directory | `scripts/battery_n400.py`, edit seed 0 | same TabMWP adapters; test split `data/distill/tabmwp/test_cases_400.jsonl` |
| FinQA targeted curriculum probe | `data/distill/poc/battery_n1_8b.json`, SHA `03dfef86ad4e`; `data/distill/poc/battery_n1_32b.json`, SHA `c808008b06b9`; text-teacher and dense-control companions in same directory | `scripts/build_n1_curriculum.py`; `scripts/run_n1_sft.sh`; `scripts/battery_n1_targeted.py`, edit seed 0 | train `data/distill/finqa/curriculum_dev_*`; probe split `data/distill/finqa/curriculum_test_strict.jsonl`; LoRA and QLoRA adapters under `data/distill/poc/lora_*_finqa_*` |
| Dense 8B control | `data/distill/poc/full_8b_chartqa/full_sft_summary.json`; `data/distill/poc/full_8b_tabmwp/full_sft_summary.json`; dense battery files under `data/distill/poc/battery_full8b_*` | `scripts/audit_full_sft_8b_nonvideo.py`; `scripts/eval_full_sft_n400.py`; `scripts/export_full_sft_8b_nonvideo_evidence.py` | full-SFT checkpoints under `data/distill/poc/full_8b_*`; summaries record trainable fraction and freeze policy |